■ 1.3.8 *Mathematica* Packages

There are a large number of functions built into the kernel of *Mathematica*. These functions provide the core of fundamental mathematical knowledge and capabilities that make *Mathematica* work. They do not, however, deal with the specific needs of every possible application area.

For many kinds of calculations, the functions that are built into *Mathematica* will be quite sufficient. However, if you are doing specialized calculations in a particular area, such as electrical engineering or particle physics, you may want to use more specialized functions. You may perhaps want a function for node voltage analysis or for Dirac gamma matrix algebra. There are no functions for these things built into *Mathematica*. However, you can always *construct* such functions from the more general ones that are built into *Mathematica*.

You will often be able to get a "package", which contains the *Mathematica* definitions for the functions you need. Each package is a file of *Mathematica* input, which you can read into *Mathematica* to "teach" it about a particular set of functions. Appendix A gives a few examples of *Mathematica* packages.

`<<`*name*	read in a *Mathematica* package

Reading in a *Mathematica* package.

This reads in a *Mathematica* package called `CombinatorialFunctions.m`.

```
In[1]:= <<CombinatorialFunctions.m
```

The function `Subfactorial` was defined in the file you just read in.

```
In[2]:= Subfactorial[10]
Out[2]= 1334961
```

Many *Mathematica* front ends provide ways for you to search through all the *Mathematica* packages on your system to try and find references to a particular topic. *Mathematica* packages are also often available in the form of notebooks, which include explanatory text, as well as *Mathematica* function definitions.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.